

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of M. Sc. (AOC) Examination November 2019

OC 724: Separation Techniques

Date: 20/11/2019 Day: Wednesday Time: 10.00 a.m. to 10.30 a.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q-I Choose the correct answer for the following questions. 20

- 1 In gas and liquid chromatography, the concentration of substance can be determine by
 - a) Comparison of the retention time of the substance with that of internal standard
 - b) Calculation of selectivity factor
 - c) Comparison of peak areas of the substance vs. that of an internal standard
 - d) Measurement of the height of the peak produced by the substance
- 2 Mass spectroscopy is used to find out
 - a) molecular weight
 - b) empirical formula
 - c) molecular formula
 - d) (a) and (b)

- 3 Which of the following condition will cause the efficiency of the column to increase?
- a) Plate number becomes greater, plate height becomes smaller
 - b) Plate number becomes smaller, plate height becomes smaller
 - c) Plate number becomes greater, plate height becomes larger
 - d) None of the above
- 4 The type of chromatography where the analyte ions in the mobile phase is attracted to the counter ions in the stationary phase is
- a) affinity chromatography
 - b) size-exclusion chromatography
 - c) ion-exchange chromatography
 - d) ion-pair chromatography
- 5 Resolution is directly proportional to
- a) cube root of N
 - b) square of N
 - c) square root of N
 - d) none of the above
- 6 In reverse phase chromatography, the stationary phase is
- a) either nonpolar or polar
 - b) non-polar
 - c) polar
 - d) none of these
- 7 Which of the following is not a common detector for gas chromatography?
- a) Refractive index detector
 - b) Thermal conductivity detector
 - c) Flame ionization detector
 - d) ECD
- 8 In DTA, following is measured
- a) ΔH vs. Temperature
 - b) ΔS vs. Temperature
 - c) ΔW vs. Temperature
 - d) ΔT vs. Temperature

- 9** In TGA, following is measured
- a) ΔT vs. Temperature
 - b) ΔW vs. Temperature
 - c) ΔH vs. Temperature
 - d) ΔS vs. Temperature
- 10** An isocratic elution in HPLC is one in which the composition of the solvent
- a) is changed continuously or in a series of steps
 - b) remains constant
 - c) both (a) and (b)
 - d) none of the above
- 11** The basis of the technique of chromatography for separating components of a mixture is
- a) the interaction of the components with a stationary and a mobile phase
 - b) the differing movement of particles of different mass in electrical field
 - c) the absorption of infrared radiation by the components
 - d) the deflection of charged particles in a magnetic field
- 12** In HPLC, the stationary phase is non-polar and a polar mobile phase is used as the eluent. Which of following would travel through the chromatography column most quickly?
- a) Iodomethane
 - b) Hexane
 - c) Chloromethane
 - d) Tetrachloromethane
- 13** Which of the following column can be used in Gas-solid chromatography?
- a) Guard column
 - b) Packed column
 - c) Analytical column
 - d) Capillary column
- 14** In DSC following is measured
- a) glass transition temperature
 - b) weight loss
 - c) decomposition temperature
 - d) none of the above

- 15** HPLC method include
- a) liquid/liquid (partition) chromatography
 - b) liquid/solid (adsorption) chromatography
 - c) ion exchange and size exclusion chromatography
 - d) all of the above
- 16** Which of the following is the most suitable gas to use as a carrier gas in GC.
- a) Carbon monoxide
 - b) Helium
 - c) Carbon dioxide
 - d) None of above
- 17** Which gas is the least suitable as a carrier gas in gas-liquid chromatography?
- a) Nitrogen
 - b) Sulphur dioxide
 - c) Helium
 - d) Argon
- 18** Which useful information can be found from Van - Deemter plot?
- a) K
 - b) Optimum column length
 - c) Longitudinal diffusion
 - d) Optimum mobile flow rate
- 19** Baseline resolution is achieved when degree of resolution/separation is
- a) 2.5
 - b) 1.5
 - c) 2.0
 - d) 0.5
- 20** In GC, the basis for separation of components is the difference in
- a) molarity
 - b) partition coefficients
 - c) molecular weight
 - d) conductivity

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Date: 20/11/2019 Day: Wednesday Time: 10.30 a.m. to 01.00 p.m. Maximum Marks: 50

Important Instructions:

- Section I and II must be attempted in TWO ANSWER SHEET.
 - Make suitable assumptions and draw neat figures wherever required.
 - Use of non-programmable calculator is allowed.
 - Show necessary calculations.
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SECTION-I

Q-II Answer the following. 10

- A** Explain Q_x and Q_y term used in relative error due to separation. 2
- B** Give the classification of chromatography based on how analyte interacts with the stationary phase. 2
- C** Explain the term involved in the following equation. 2

$$T_{n,r} = \frac{n!}{r! (n-r)!} \left[\frac{1}{K+1} \right]^n \cdot K^r$$

- D** Write note on: Gas Density Balance Detector (GDBD). 4

OR

- D** Write note on: Electron Capture Detector (ECD). 4

Q-III Answer the following. 10

- A** Explain C term of Van-Deemter equation in GC. 2
- B** What is artifact peak and explain the causes of artifact peak. 2
- C** Give the method to determine carbon in H_2O_2 by combustion gas chromatography. 3

OR

- C** Mention the characteristics of the detector used in GC. 3
- D** Draw the typical chromatogram and explain it. 3

SECTION-II

Q-IV	Answer the following.	10
A	Mention advantages and disadvantages of the RI detector.	2
B	What are the factors affecting eddy diffusion in HPLC ?	2
C	Explain: Reverse Phase Chromatography.	3
D	Draw the block diagram of GC and explain it.	3
OR		
D	Mention advantages and limitation of HPLC.	3
Q-V	Answer the following.	10
A	Explain: Isocratic analysis in HPLC.	2
B	Draw a block diagram of TG and DSC.	3
C	Write note on light scattering detector.	3
D	Draw the block diagram of HPLC.	2
Q-VI	Answer the following.	10
A	Mention the applications of DSC.	3
B	Give the factor affecting TG curve.	3
C	Show the formation of di-cation in naphthalene molecular ion.	4
OR		
C	The mass spectra data can distinguish between 1-propanol and 2-propanol. Justify the statement.	4